

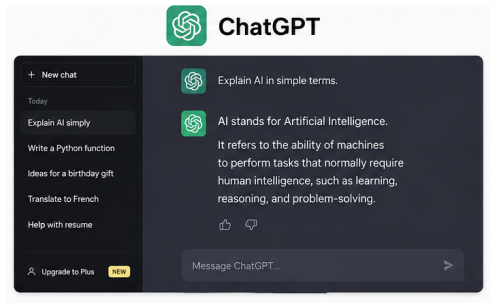
AI & social networks

when IA reproduces and amplifies inequality

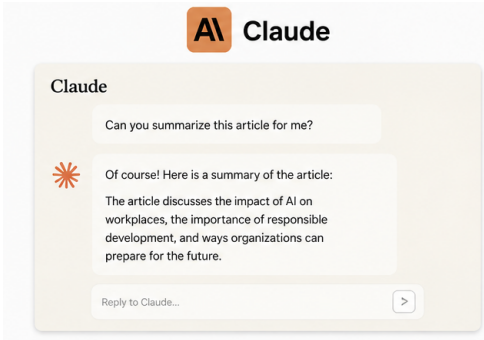
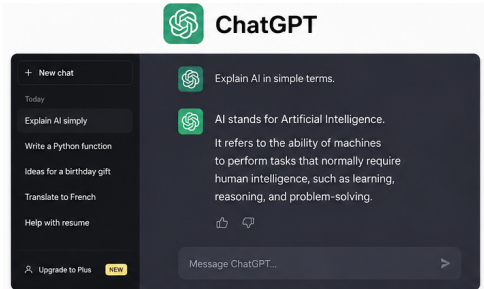
Mathilde Perez, Télécom Paris, IP Paris



Where is AI ?



Where is AI ?

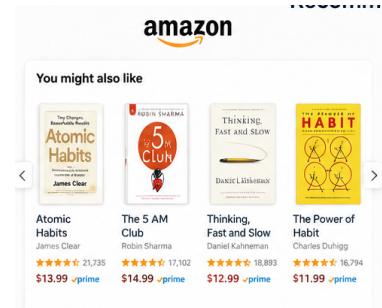
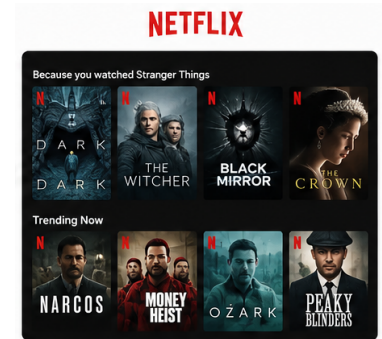
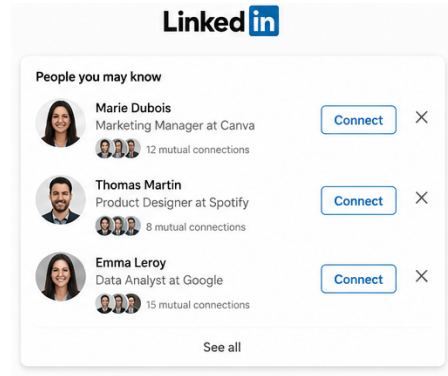


Recommender algorithm

what you see

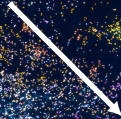
who you should connect with

what you might buy



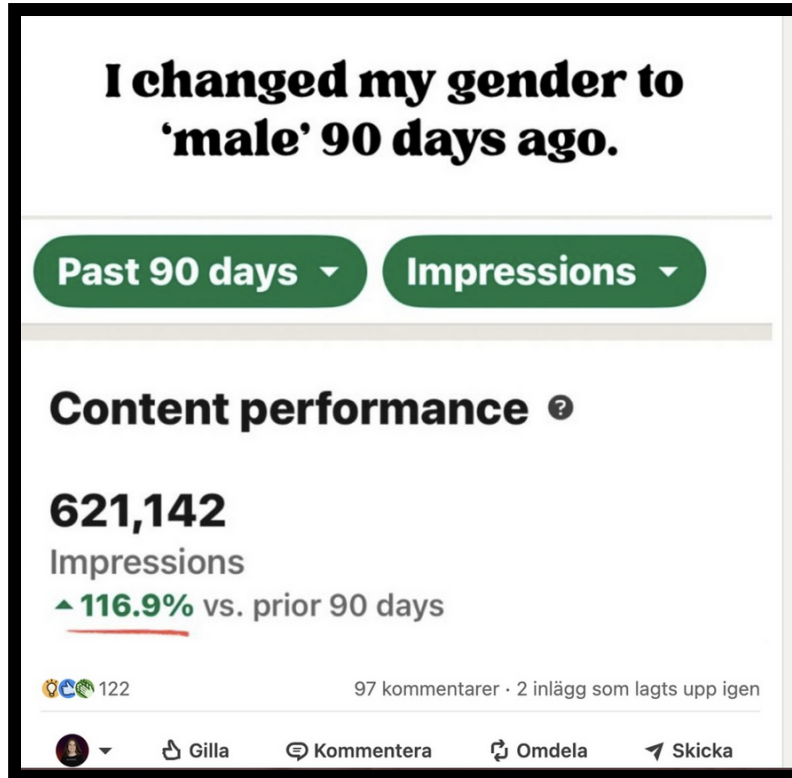


Why I'm here?



What could go wrong?

What could go wrong?



Why should companies care?

- ▶ Entered into force in **2024**
- ▶ Classifies recommendation algorithms as **high-risk** for some applications
- ▶ Requires **transparency, accountability, and fairness**



Fairness goes from **academic concern** to **legal obligation**

What does it mean to be fair?

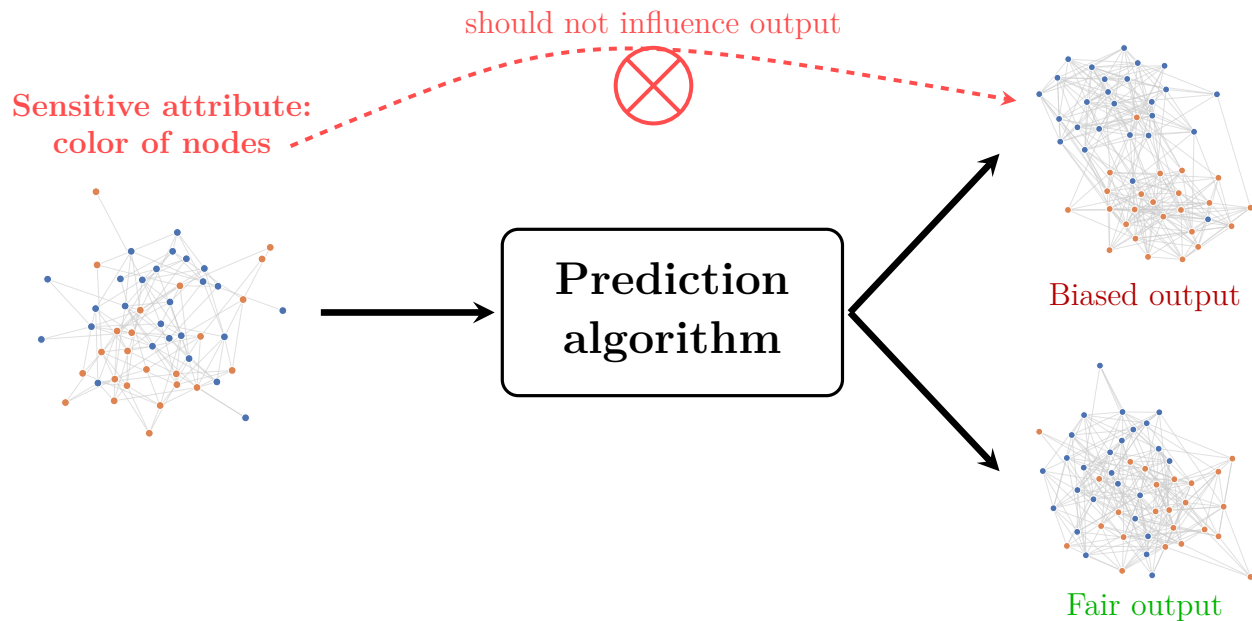


In Machine Learning, **fairness** means that a model's predictions should not be influenced by **sensitive attributes** :

- **Gender**
- **Race / Ethnicity**
- **Age**
- **Religion, origin...**

A model should make the **same decision** regardless of who you are, based only on **relevant information**.

e.g. a hiring algorithm should not reject a candidate because she is a woman.



- ▶ **Bias lives in data** : historical data reflects past inequalities
- ▶ **The goal** : learn useful patterns without reproducing discriminatory ones
- ▶ **The challenge** : bias is structural, not just a column to remove

How to remove the sensitive attribute : Woman ?

Mathilde Perez

PhD in Fairness



- ▶ **Age** : 27
- ▶ **Gender** : woman
- ▶ **Country** : French
- ▶ **Hobbies** : Women club
chess
- ▶ **Education** : Mathematics
University
- ▶ **Current position** : PhD
student, Télécom Paris

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Clearly not enough..

- ▶ **Age** : 27
- ▶ **Gender** : ~~woman~~
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Always not..

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How to remove the sensitive attribute : woman ?

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It can still be guessed...

- ▶ **Age** : 27
- ▶ **Gender** : ~~woman~~
- ▶ **Country** : French
- ▶ **Hobbies** : Women club chess
- ▶ **Education** : Mathematics University
- ▶ **Current position** : PhD student, Télécom Paris

How to remove the sensitive attribute : woman ?

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PhD in Fairness



It's starting to get a lot...

- ▶ **Age** : 27
- ▶ **Gender** : ~~woman~~
- ▶ **Country** : French
- ▶ **Hobbies** : ~~Women-club~~
~~chess~~
- ▶ **Education** : Mathematics
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No we can't do that.

- ▶ **Age** : 27
- ▶ **Gender** : ~~woman~~
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~~chess~~
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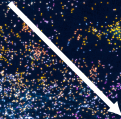
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We have to delete some
connection also

→ we loose too much information.

Why I'm here?



My Research

*What is the influence of recommendation systems on network evolution?
How will networks evolve in the future?*

Homophily = $\underbrace{\text{Choice}}_{\substack{\text{intrinsic preferences} \\ \text{Who I want to connect to}}} + \underbrace{\text{Induced}}_{\substack{\text{exposure \& structural constraints} \\ \text{Who I am exposed to}}}$

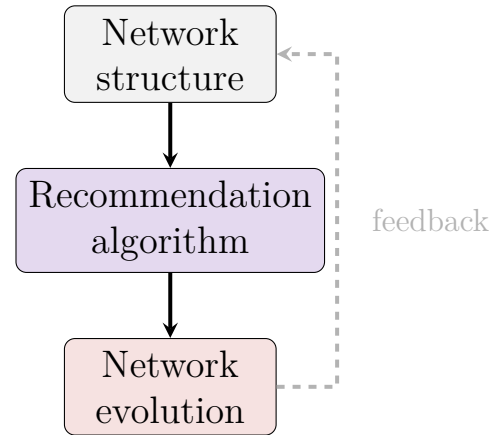
- ▶ **Recommendation systems** shape exposure and therefore homophily
- ▶ **Feedback loop** : the network influences recommendations, which influence the network
- ▶ **Key question** : can we disentangle what is chosen from what is imposed?

My Approach

Mathematical modelling of networks & dynamics

- ▶ **Graph models** : represent the network structure and its evolution over time
- ▶ **Dynamic models** : capture how connections form, break, and reinforce
- ▶ **Recommendation as a perturbation** : inject algorithmic exposure into the model

→ Measure **how much** and **how fast** recommendation algorithms shape the network



Thank you!

